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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Sc DEGREE EXAMINATION APRIL 2021

ASCS301Q – STATISTICAL METHODS FOR COMPUTER APPLICATIONS - I

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2 =20)

Answer ALL the questions

1. What is classification?.
2. What is the use of ogive curve?
3. Define measure of location.
4. State the difference between absolute and relative measure of dispersion.
5. Write the formula used to calculate Bowley's co.efficient of skewness?
6. Define Skewness.
7. Define correlation.
8. Write the two regression equations.
9. What is the principle of least square?
10. Write the two normal equations for fitting straight line.

SECTION-B (5×5=25)

Answer ALL the questions

11. a. Explain the chronological classification with an example.

(Or)

b. Explain the different draw backs of Statistics.

12. a. Explain the properties of standard deviation.

(Or)

b. Calculate the mean deviation from the mean for the following data.

Size:	2	4	6	8	10	12	14	16
Frequency:	2	2	4	5	3	2	1	1

13. a. A frequency distribution showed the following measure of location:

Mean =45, Median =48 , co.efficient of skewness= - 0.4.

Find the value of Standard deviation by using the above information.

(Or)

b. Find the co.efficient of skewness ,if
Difference between two quartiles=8
Sum of two quartiles=22
Median=10.5

14. a. Explain about scatter diagram.

(Or)

b. Calculate the co.eficient of correlation from te following data by the Spearman's Rank difference method:

X:	75	88	95	70	60	80	81	50
Y:	120	134	150	115	110	140	142	100

15. a Explain the method of obtaining normal equations for second degree polynomial.

(Or)

b. Explain the method of fitting power curve.

SECTION - C (3×10=30)

Answer any THREE Questions

16. Draw Histogram and find the value of mode from the following data.

Marks:	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Freq.:	5	15	30	40	60	40	30	15	5

17. Calculate the value of Arithmetic Mean and Median from the following table.

Class:	10-14	14-18	18-22	22-26	26-30	30-34	34-38	38-42	42-46
Freq:	2	4	4	8	12	16	10	8	4

18. Compute Karl-Pearson's co.efficient of skewness from the following data.

Profits:	10-20	20-30	30-40	40-50	50-60
No. of companies:	18	20	30	22	10

19. Obtain the two regression equations from the following data.

X:	60	62	65	70	72	48	53	73	65	82
Y:	68	60	62	80	85	40	52	62	60	81

20. Fit a straight line by the method of least squares from the following data.

X:	6	2	10	4	8
Y:	9	11	5	8	7

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